Assignment 3 Report

Data Science for Software Engineering

Brian Stevo Aranha
Achara Obinna Vincent
Namit Joshi
Anil Kumar

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1 Introduction

1.1 Introduction

Issue tracking systems such as Jira contain useful software engineering discussions about bugs, improvements, features and design decisions. Prior work has shown that bug reports provide important information for developers, although the quality and completeness of this information can vary [2, Introduction]. In this task, we focus on Apache Tika issues from the Apache Jira issue tracker [1]. The aim of Week 1 was to prepare the issue data for later topic modeling.

2 Study Design

We used the `Tika` sheet from the provided Excel file because Apache Tika is our selected project. The sheet contained 211 issue IDs and three design-decision labels, existence, property and executive.

Using Python, we downloaded the corresponding issue data from Apache Jira. The downloaded fields included summary, description, issue type, status, resolution, priority, labels, components, dates, parent issue, number of comments and number of attachments.

For preprocessing, we combined each issue summary and description into one text field. The text was then cleaned by lowercasing, removing URLs and special characters, splitting camel-case words, removing stop words, lemmatizing tokens and removing very short tokens.

3 Results

3.1 Week 1: issue Data Download and Vocabulary Creation

A total of 211 Apache Tika issues were processed. All issues were successfully downloaded from Jira using the provided issue list. For each issue, the summary, description, issue type, status, priority, parent issue information, number of comments and number of attachments were collected.

After downloading the data, the issue summary and description were combined and preprocessed. After preprocessing, no issue had zero tokens. The average number of tokens per issue was 69.78.

The vocabulary contained 14,724 total tokens and 2,706 unique tokens. The most frequent tokens included `parser`, `file`, `metadata`, `type`, `text`, `content`, `test`, `code`, `document`, `version`, `pdf`, `user`, `server`, `stream`, `data`, `parse`, `class`, `java`, `vulnerability` and `cve`.

Finally, a document-term matrix was created from the preprocessed issue text. The matrix contains 211 issues and 2,706 token features. With the metadata columns included, the final matrix size is 211×2710. This matrix will be used as input for LDA topic modeling in the next stage of the assignment.

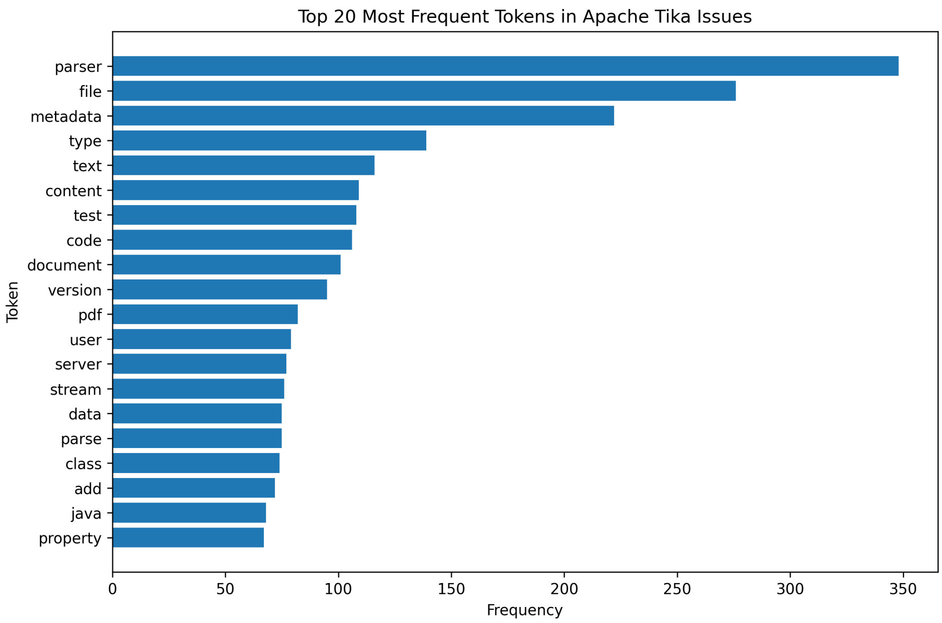

Figure 3.1: Top 20 most frequent tokens in the preprocessed Apache Tika issue texts.

3.2 Week 2:

3.3 Week 3:

4 Discussion

The Week 1 preparation was successful because all 211 issues were downloaded and all issues still had usable text after preprocessing. The most frequent tokens reflect the Apache Tika domain. For example, terms such as `parser`, `file`, `metadata`, `document`, `pdf` and `content` were expected because Tika is used for parsing files and extracting text and metadata.

Some tokens also suggest possible future topics. For example, `vulnerability`, `cve` and `vulnerable` may form a security-related topic.

5 Threats to Validity

5.1 Internal Validity

The preprocessing choices may affect the later topic modeling results. For example, removing stop words, lemmatizing tokens, or deleting short words can change the final vocabulary.

5.2 External Validity

The study focuses only on Apache Tika issues. Therefore, the results may not generalize to other software projects because each project has its own terminology and issue discussions.

5.3 Construct Validity

Only the issue summary and description were used for text preprocessing in Week 1. Some useful design information may also exist in comments or attachments, but these were not included in the text analysis at this stage.

6 Conclusion and Future Work

In Week 1, we prepared the Apache Tika issue dataset for topic modeling. We cleaned the issue list, downloaded Jira issue data, preprocessed the issue summary and description, created a vocabulary and generated the document-term matrix.

A Team Effort and Contributions

In Table A.1, we detail the estimated hours and main activities completed by each team member throughout the assignment.

Table A.1: Summary of team effort in Week 1

Week 1		
Name	**Hours**	**Activities**
Brian	0	•
Obinna	6	• Organized the Week 1 assignment folder and project files. • Processed the Apache Tika issue list and downloaded the corresponding Jira issue data. • Preprocessed the issue summaries and descriptions, then created the vocabulary and document-term matrix.
Namit	0	• •
Anil	8	• Fetched all issues from the Apache Jira API across projects • Summary and description per issue, then applied tokenization, stop word removal, lemmatization, and ontology class replacement • Created a filtered vocabulary from all unique tokens

Table A.2: Summary of team effort in Week 2

Week 2		
Name	**Hours**	**Activities**
Brian	0	• •
Obinna	0	• •
Namit	0	• •

Table A.3: Summary of team effort in Week 3

Week 3		
Name	**Hours**	**Activities**
Brian	0	• •
Obinna	0	• •
Namit	0	• •

Bibliography

[1] Apache Software Foundation. *Apache Tika Issues on Apache Jira*. `https://issues.apache.org/jira/projects/TIKA/issues`. Issue tracking data, accessed 26 June 2026. 2026.

[2] Nicolas Bettenburg et al. “What Makes a Good Bug Report?” In: *Proceedings of the 16th ACM SIGSOFT International Symposium on Foundations of Software Engineering*. SIGSOFT ’08/FSE-16. ACM, 2008, pp. 308–318. DOI: `10.1145/1453101.1453146`.